

Your grade: 100%

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Next item →

1.

What property does the general Bayesian recursion integral framework derived this week share with the Kalman filter derived in the second course in this specialization?

1 / 1 point

☒ The solution for both derivations breaks up into prediction and updating computations for every iteration.

☐ Both methods assume that the process and sensor noises have Gaussian distributions.

☐ Both methods assume that the state vector has a Gaussian distribution.

☐ Both methods seek to approximate the conditional mean of the state vector, $\mathbb{E}[x_k \mid \mathcal{Z}_k]$.

☒ Correct

Yes, this is correct.
2.

If we know $f(x_k \mid \mathcal{Z}_k)$, what metrics can we compute (in principle, although it may sometimes be difficult to do so in practice)? Select all that apply.

1 / 1 point

☒ The conditional mean of x_k given $\mathcal{Z}_k$, $\mathbb{E}[x_k \mid \mathcal{Z}_k]$.

☒ The conditional variance (or covariance) of x_k given $\mathcal{Z}_k$, $\text{var}(x_k \mid \mathcal{Z}_k)$.

☒ The conditional mode of x_k given $\mathcal{Z}_k$.

☒ The conditional median of x_k given $\mathcal{Z}_k$.

☒ Correct

Yes, this is correct.
3.

We seek to approximate $\int_0^4 x^3 dx$ using the rectangular rule, with $w = 1$. For convenience, the data you will need to perform this computation are presented in the table:

1 / 1 point

x	$g(x) = x^3$
0.5	0.1250
1.5	3.3750
2.5	15.6250
3.5	42.8750

What is the approximate integral computed by the rectangular rule? Enter your answer with two digits to the right of the decimal point.

62

☒ Correct

Yes, this is correct.
4.

We seek to approximate $\int_0^4 x^3 dx$ using the trapezoid rule, with $w = 1$. For convenience, the data you will need to perform this computation are presented in the table:

1 / 1 point

x	$g(x) = x^3$
0	0
1	1
2	8
3	27
4	64

What is the approximate integral computed by the trapezoid rule? Enter your answer with two digits to the right of the decimal point.

68

☒ Correct

Yes, this is correct.
5.

Suppose that we wish to track the 1-dimensional state of a system modeled by:
 $x_{k+1} = \cos(x_k^2) + w_k$
 $z_k = x_k^2 + v_k$,
where $w_k \sim \mathcal{N}(0, \sigma_w^2)$ and $v_k \sim \mathcal{N}(0, \sigma_v^2)$. What is $f(x_k \mid x_{k-1})$?

1 / 1 point

☐ $\mathcal{N}(0, \sigma_w^2)$.

☐ $\mathcal{N}(\alpha x_{k-1}, \sigma_w^2)$.

☒ $\mathcal{N}(\cos(x_{k-1}^2), \sigma_w^2)$.

☐ $\mathcal{N}(0, \sigma_v^2)$.

☐ $\mathcal{N}(x_k^2, \sigma_v^2)$.

☐ $\mathcal{N}(\sin(x_k), \sigma_v^2)$.

☒ Correct

Yes, this is correct.
6.

Suppose that we wish to track the 1-dimensional state of a system modeled by:
 $x_{k+1} = \cos(x_k^2) + w_k$
 $z_k = x_k^2 + v_k$,
where $w_k \sim \mathcal{N}(0, \sigma_w^2)$ and $v_k \sim \mathcal{N}(0, \sigma_v^2)$. What is $f(z_k \mid x_k)$?

1 / 1 point

☐ $\mathcal{N}(0, \sigma_w^2)$.

☐ $\mathcal{N}(\alpha x_{k-1}, \sigma_w^2)$.

☐ $\mathcal{N}(\cos(x_{k-1}^2), \sigma_w^2)$.

☐ $\mathcal{N}(0, \sigma_v^2)$.

☐ $\mathcal{N}(0.1x_k^2, \sigma_v^2)$.

☒ $\mathcal{N}(x_k^2, \sigma_v^2)$.

☒ Correct

Yes, this is correct.
7.

How do we compute $f(z_k \mid \mathcal{Z}_{k-1})$ in the denominator of the time-update equation for $f(x_k \mid \mathcal{Z}_k)$?

1 / 1 point

☐ We apply the law of total probability and write this as an integral that we can approximate using either the rectangular rule or the trapezoid rule.

☒ We don't, it is needed only as a normalization factor that we can determine in other ways.

☒ Correct

Yes, this is the approach that we will take.
8.

Execute the Coursera Lab Jupyter notebook for Lesson 4.1.6, which implements the brute-force numeric approximation to Bayesian inference.

Use the default model and make sure that the code executes `randn("state",0)` and `rand("state",0)` and `randg("state",0)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

Set `nR = 100`; then, run the code. What is the root-mean-squared error between the true state and the estimate of the **mean** produced by the code? Enter your result with three digits to the right of the decimal point.

1 / 1 point

0.295

☒ Correct

Yes, that is correct.
9.

Execute the Coursera Lab Jupyter notebook for Lesson 4.1.6, which implements the brute-force numeric approximation to Bayesian inference.

Use the default model and make sure that the code executes `randn("state",0)` and `rand("state",0)` and `randg("state",0)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

Set `nR = 300`; then, run the code. What is the root-mean-squared error between the true state and the estimate of the **mode** produced by the code? Enter your result with three digits to the right of the decimal point.

1 / 1 point

0.314

☒ Correct

Yes. That is correct.
10.

Execute the Coursera Lab Jupyter notebook for Lesson 4.1.6, which implements the brute-force numeric approximation to Bayesian inference.

Use the default model and make sure that the code executes `randn("state",0)` and `rand("state",0)` and `randg("state",0)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

Set `nR = 300`; then, run the code. What is the root-mean-squared error between the true state and the estimate of the **median** produced by the code? Enter your result with three digits to the right of the decimal point.

1 / 1 point

0.293

☒ Correct

Yes, that is correct.